

Effect of X-ray spectrum on size-specific dose estimates (SSDEs) for paediatric and adult patients undergoing CT scans

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ABSTRACT

The size-specific dose estimate (SSDE) concept is based on converting volume averaged CT dose index (CTDI_{vol}) measured in cylindrical PMMA head and body phantoms of reference sizes to values for doses absorbed in individual patients. The original conversion factors used for SSDE is related to the effective diameter (ED) of the region of interest for the patient. ED is simple to measure, but does not include attenuation, which affects estimation of the dose absorbed. Water-equivalent diameter (D_w) which takes account of attenuation through the CT number, based on the linear attenuation coefficients, has been recommended to replace ED. The main scan parameter that affects D_w values is the x-ray spectrum. In this study, a library of 363 phantoms (168 paediatric and 195 adult) of a wide range of sizes from newborn to large adult were used to investigate the influence of the x-ray spectrum on D_w and the correlation between ED and D_w. Values were assessed for each phantom over seven scanned regions: head, chest, abdomen, pelvis, chest-abdomen, abdomen-pelvis, and the whole trunk using x-ray spectra of 80–140 kVp. The results show the effect of x-ray spectrum on the relationship between D_w and ED for the trunk regions was minimal, and almost all D_w/ED ratios were within a factor of ± 1.01 . However, D_w/ED values for the head region varied by a factor of up to 1.10 with the x-ray spectra. Energy-independent correlations between ED and D_w values for the trunk regions have been established. The correlations were compared with published results calculated by AAPM using another method and gave an average difference of 2.9%. Use of the correlations should simplify determination of the SSDE taking account of attenuation based on ED that is more readily measurable than D_w, which requires advanced software.

1. Introduction

Radiation exposures resulting from diagnostic medical procedures are reviewed continually to ensure they are as low as is appropriate while providing the diagnostic information required. A trade-off between benefits and risks of medical procedures is considered in optimization of doses delivered to patients. Computed tomography (CT) is an imaging modality, that is widely used, and because the doses involved are relatively large, it contributes a high proportion of the collective effective dose from medical exposures (AAPM, 2008; Kalender, 2014). Therefore, attention has been paid in the last two decades to methods for minimizing CT scan doses as much as possible (Kalender, 2006, 2014; Shrimpton et al., 2015; Tonkopi et al., 2017; AAPM, 2019a).

Doses absorbed by patients undergoing CT scans are estimated using two indices, the volume averaged CT dose index (CTDI_{vol}) and the dose-

length product (DLP) (AAPM, 2008). Measurements of the CTDI_{vol} are performed with two cylindrical PMMA phantoms of reference sizes that represent average sizes for the adult head and body. Diameters of these phantoms are 16 cm and 32 cm, and both are 15 cm in length. The small phantom is also used for dose measurements of the paediatric body, but the current recommendation is to perform the measurements with the large phantom for all paediatric and adult body scans and only use the small one for head scans (HERCA, 2017). This is to avoid any confusion among health practitioners and scanner manufacturers, where some manufacturers use the small phantom for paediatric body, and other manufacturers use the large one. The DLP is derived by multiplying CTDI_{vol} by the length of the scan. Both indices are displayed prior to the scan and given in the patient dose report after the scan, which can be utilized retrospectively to review a patient dose using, for example, dose tracking software that has become popular in some countries for dose optimization. They are also used to compare scan protocols and

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scanner performance as well as establishing national or regional diagnostic references levels (DRLs) (ICRP, 2017).

Since $CTDI_{vol}$ is measured in standard phantoms of reference sizes, it only gives an estimate for the dose delivered to the tissues of patients similar in size to the phantom. This means that $CTDI_{vol}$ and DLP do not provide accurate estimates of doses for individual patients. In order to address this, a size-specific dose estimate (SSDE) concept has been developed to provide a dose index related more closely to the doses to tissues for individual patients linked to their size, but based on $CTDI_{vol}$ (AAPM, 2011). SSDE depends on applying a size conversion factor to account for the size difference between the patient and the reference phantom. The conversion factors have been developed through collaboration between various independent research groups and provided in the AAPM reports of task groups (TGs) 204 and 220 for body scans and TG 293 for head scans (AAPM, 2011, 2014, 2019b).

An effective diameter (ED) assessed from the lateral and anterior-posterior dimensions for the scanned region of the patient is used as a measure of patient size (AAPM, 2011), but ED cannot distinguish between regions of low or high attenuation. The lack of data on attenuation can lead to over or underestimation of the patient dose (Wang, Christner, et al., 2012; Wang, Duan, et al., 2012). For example, the size of the thoracic region is comparable to that for other regions of the body, but its attenuation is lower because the lungs that have a low density occupy a large portion of the space. Therefore, it has been suggested that patient size is assessed in terms of water-equivalent diameter (D_w) taking account of the attenuation within the scanned region to overcome this limitation (AAPM, 2014, 2019b). D_w will not only vary for different regions of the body, but will be influenced by scan protocols and parameters used for scans of individual patients. Thus, D_w for the same region of the body will vary depending on scan parameters, whereas ED would be constant under all conditions.

Since the D_w value is based on attenuation of the scanned region, the main scan parameter that would have an impact on its value is the x-ray spectrum, represented by the applied tube potential (kVp). The aims of this study are (1) to investigate the influence of the scan x-ray spectrum on D_w values for paediatric and adult patients, and (2) to study the correlation between ED and D_w values under different conditions.

2. Materials and methods

2.1. The SSDE concept

The SSDE concept involves estimating the doses absorbed by patients of specific sizes for a given CT scan using the $CTDI_{vol}$, which is measured periodically in the standard phantoms to ensure stability of x-ray tube performance. The SSDE converts the CTDI value to that appropriate for a patient of a specific size using a conversion factor f_{size} as follows:

$$\begin{aligned} SSDE &= f_{size}^{16cm} \times CTDI_{vol,16cm} \\ SSDE &= f_{size}^{32cm} \times CTDI_{vol,32cm} \end{aligned} \quad (1)$$

where 16 and 32 indicate the standard head or body CTDI phantoms, respectively. The CTDI phantom used for $CTDI_{vol}$ must be known to assess SSDE because the conversion factors differ for each phantom. Thus, 16 cm and 32 cm symbols are usually used with the conversion factor to distinguish between the head (f_{size}^{16cm}) and body (f_{size}^{32cm}) factors. The conversion factor for head or body scans can be determined for a specific patient by the patient size D_w as follows:

$$\begin{aligned} f_{size}^{16cm} &= a_{16cm} \times e^{-b_{16cm} \times D_w} \\ f_{size}^{32cm} &= a_{32cm} \times e^{-b_{32cm} \times D_w} \end{aligned} \quad (2)$$

where a and b are the correlation coefficients between f_{size} and D_w for the head (16 cm) and body (32 cm). These correlations are reported in

the AAPM reports for each phantom size (AAPM, 2014, 2019b).

2.2. The patient size methods

The ED is based on the anterior-posterior dimension (AP) and the lateral dimension (LAT) of the patient's cross-sectional area in the middle of the scanned region. It represents patient size as diameter of a circle that has an area similar to that of the patient's cross-section as:

$$ED = \sqrt{AP \times LAT} \quad (3)$$

The water-equivalent diameter, on the other hand, describes the patient size in terms of a cylindrical water phantom that would absorb a dose similar to the scanned region of the patient. This approach considers variations in tissue densities and attenuations inside the scanned area. Similar to ED, calculation of D_w can be based on the CT scan projection radiograph (SPR) that is taken before the scan or on the reconstructed CT images. Although it is desirable to estimate D_w and hence SSDE before a scan is performed to optimize patient dose and image quality, there are practical issues (AAPM, 2014). For instance, the pixel values of the SPR image need to be calibrated in terms of water attenuation, but this calibration depends on the scanner and varies among manufacturers. Thus, it is recommended that this calibration is performed by the scanner manufacturers.

Since the SPR image differs from the CT image, the method used to calculate D_w is different in the two approaches, whereas the methods used to derive ED are similar because both methods, only depend on the dimensions. In this study, the method used for CT images is considered for D_w . This method involves measuring the water-equivalent area (A_w) of each slice of the scanned region, i.e. the axial images. The area measured only includes the patient body, and other objects that are not related to the patient, such as the scanner table, are removed. This is accomplished by segmentation of the patient body within each slice. The segmented area is called the region of interest (ROI). Assessment of A_w is based on the CT numbers (CT (x,y)) of the scanned region pixels, which are typically reported in Hounsfield Units (HU), and determined by the linear attenuation coefficient of each pixel of the scanned region as:

$$CT(x, y) = \left(\frac{\mu_{(x,y)} - \mu_{water}}{\mu_{water}} \right) \times 1000 \quad (4)$$

where $\mu_{(x,y)}$ is the linear attenuation coefficient of a tissue located in the pixel (x,y) at a given photon energy normalized to that of water μ_{water} at the same energy.

A_w is assessed by means of the CT numbers over the (ROI) as:

$$A_w = \frac{1}{1000} \times \overline{CT(x, y)}_{ROI} \times A_{ROI} + A_{ROI} \quad (5)$$

where A_{ROI} is the sum of the pixel areas ($A_{ROI} = \sum A_{pixel}$) of the ROI. Subsequently, D_w of each slice can be calculated as:

$$\begin{aligned} D_w &= 2 \times \sqrt{\frac{A_w}{\pi}} \\ D_w &= 2 \times \sqrt{\left(\frac{1}{1000} \times \overline{CT(x, y)}_{ROI} + 1 \right) \times \frac{A_{ROI}}{\pi}} \end{aligned} \quad (6)$$

2.3. Assessment of ED and D_w

A MATLAB code developed in house was employed to estimate values of ED and D_w for a library of 363 paediatric and adult phantoms reconstructed from CT images of more than 1000 CT scans of a wide range of ages and sizes. The library is known as PHANTOMS (NCI, 2020), and was developed by the University of Florida and the National Cancer Institute in three dimensions to represent the human body (Lee et al., 2010, 2013). The phantoms were created with various ages, lengths, and weights for both genders so that the collection includes a full range of sizes for the patient population. It includes 168 paediatric

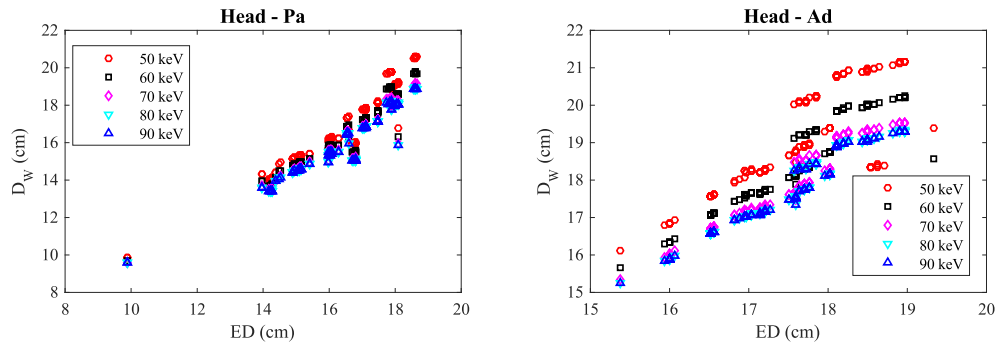


Fig. 1. D_w values as a function of ED calculated with various mean energies for the paediatric (Pa) and adult (Ad) phantoms in the head region.

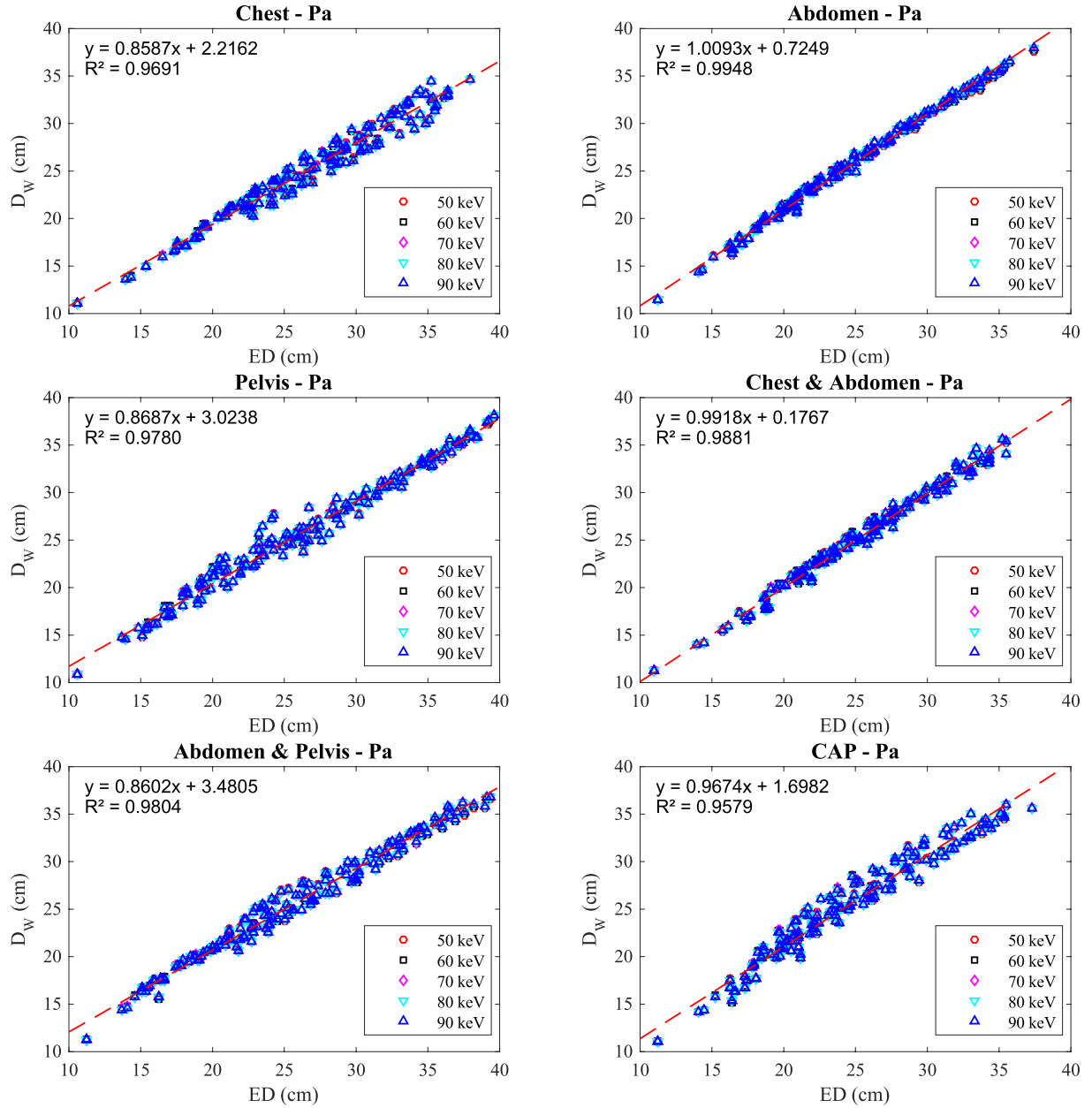


Fig. 2. D_w values as a function of ED calculated with various mean energies for the paediatric (Pa) phantoms in the trunk regions. The red dashed line represents the linear regression for the correlation between the values. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

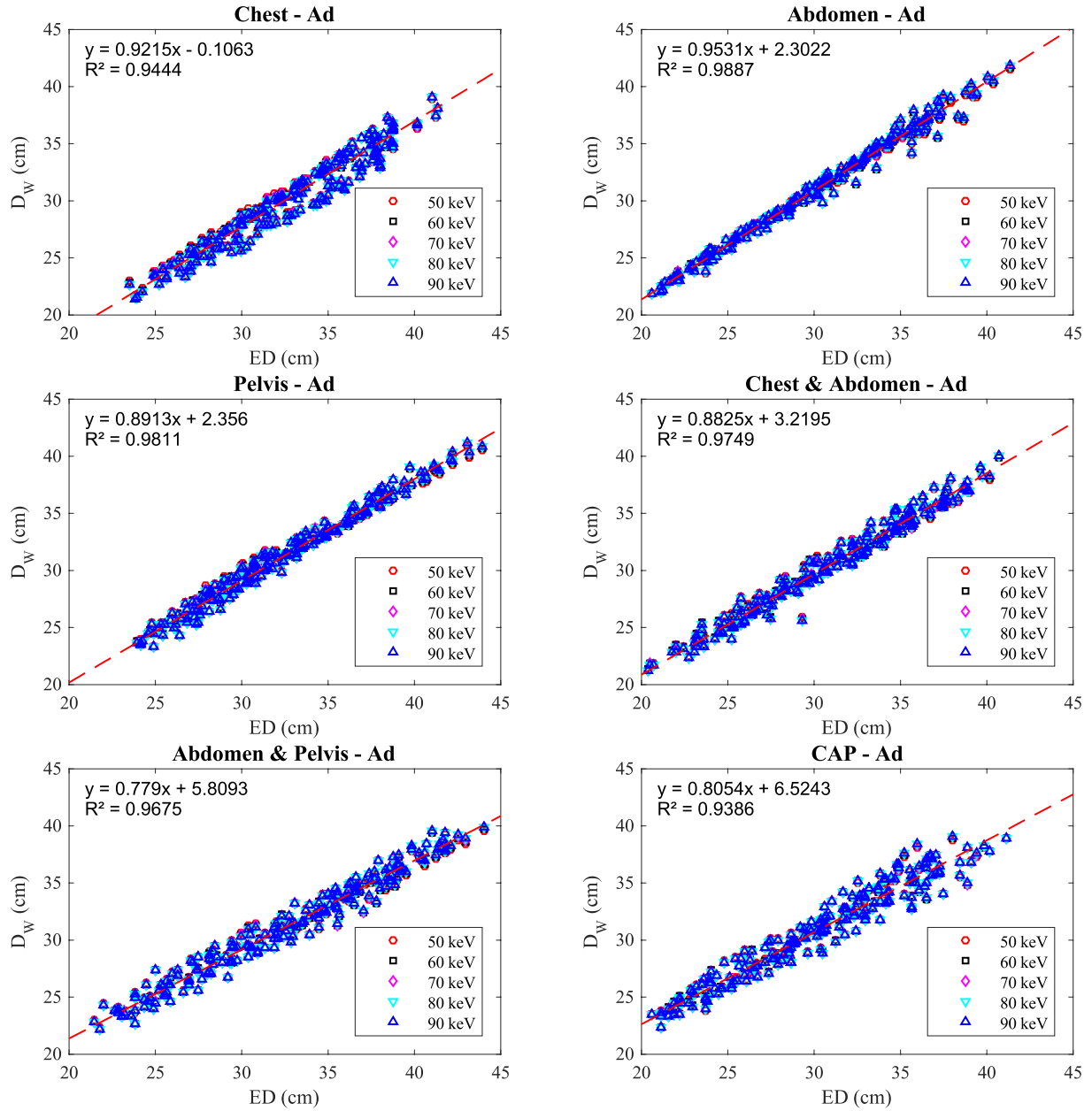


Fig. 3. D_w values as a function of ED calculated with various mean energies for the adult (Ad) phantoms in the trunk regions. The red dashed line represents the linear regression for the correlation between the values. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

phantoms ranging from a small baby size to 15 years old, and lengths and weights in the range of 51–175 cm and 4–115 kg, respectively. It also contains 195 adult phantoms with length and size ranges of 150–190 cm and 40–125 kg. Moreover, 12 phantoms (10 paediatric and 2 adults) from the library were developed with lengths and weights that represent reference paediatric and adult patients that are recommended by the International Commission on Radiological Protection (ICRP) (ICRP, 2009, 2020).

The head region and six scanned regions similar to those used in a recent study were investigated (Abuhaimed and Martin, 2020). The regions were head, chest, abdomen, pelvis, chest and abdomen, abdomen and pelvis, and the whole trunk (CAP). The ED values of each phantom were assessed as in Eq. (3) in the middle position of the seven scanned regions studied (AAPM, 2011). D_w was assessed for each slice $D_{w,slice}$ of the regions studied as in Eq. (6), and the mean was calculated as (AAPM, 2014):

$$D_w = \sum_{z=1}^N \frac{D_{w,slice}(z)}{N} \quad (7)$$

where ($Z = 1$) is the first slice of the region, and N is slices number of the region.

2.4. Influence of x-ray spectrum

As mentioned previously, the x-ray spectrum used for a scan would affect values of D_w , but not ED. In order to determine how significant the influence is, a range of x-ray spectra was used to assess D_w for each region. D_w assessment is based on the $CT(x, y)$ of each pixel (Eq. (6)), which is determined by values of $\mu_{(x,y)}$ and μ_{water} at the tube potential applied (Eq. (4)). The manual calculation of D_w for a spectrum is difficult, as $CT(x, y)$ values vary as the photon energy distribution changes through the patient cross-section with beam hardening, and this also

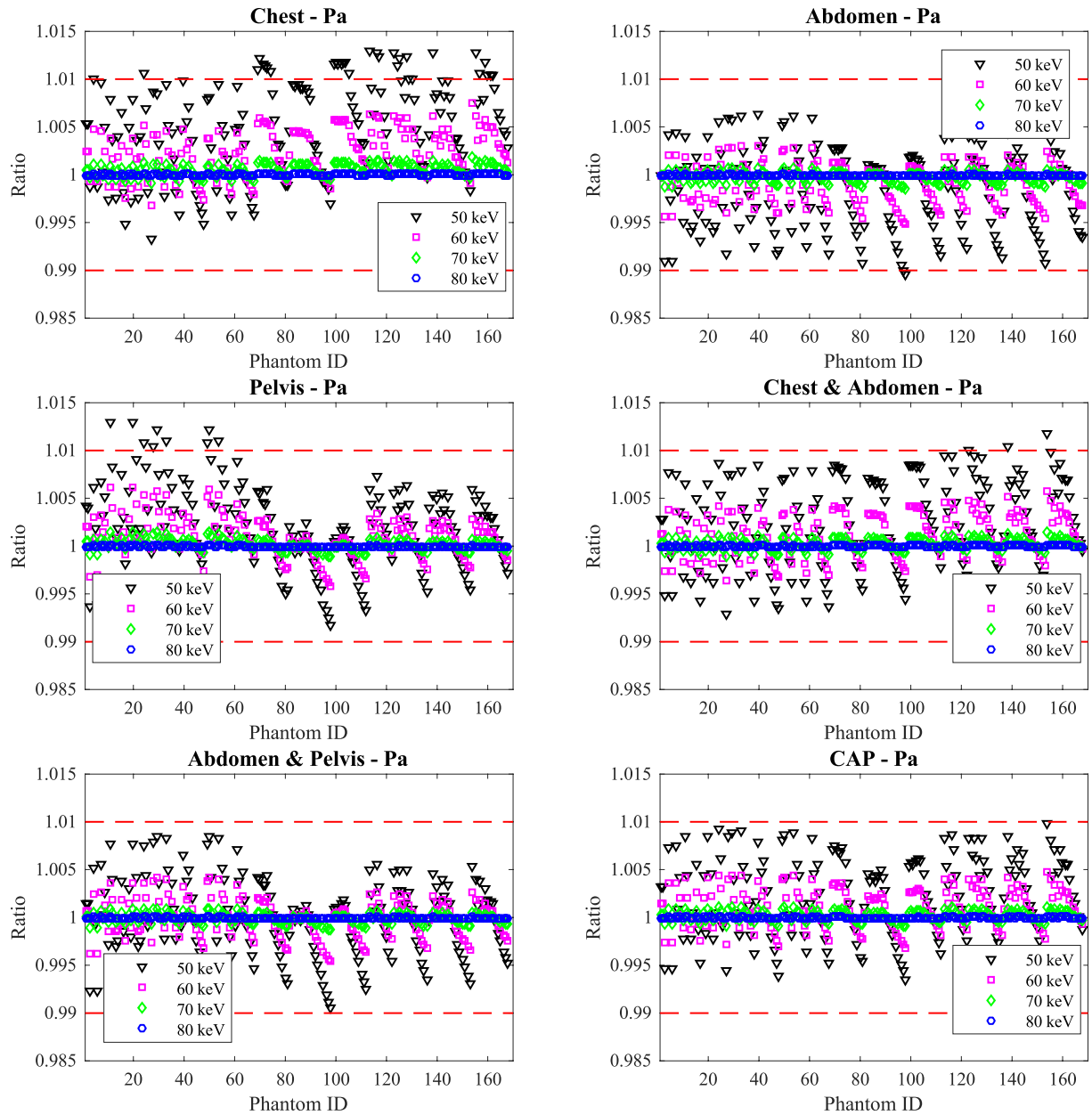


Fig. 4. Ratios of D_w calculated for the six regions of the trunk for the paediatric (Pa) phantoms with various mean energies and normalized with respect to those for 90 keV.

depends on the size of the patient. $CT(x, y)$ values will differ for patients undergoing the same scan protocol and with the same scanning parameters. However, D_w can be estimated based on the mean value of $\overline{CT}(x, y)$ for the ROI as in Eq. (6).

In order to investigate the influence of the x-ray spectrum on the patient size accurately, the patient size should be held constant to eliminate all differences, while changing the tube potential from 80 to 140 kVp, which is the typical tube potential range applied to perform CT scans. This would be possible if actual images are used, but some uncertainties would be expected, as even if data from two patients of similar sizes were available, the body structures and positions of organs and tissues will be different. D_w values were calculated for the same phantoms in a previous study using an effective energy for a spectrum of 120 kV that was close to what had been used in (Leng et al., 2015), where their study was based on using CT numbers of actual patient images for adults. It has been found that D_w values calculated using the effective energy were in a good agreement with those based on CT

images (Abuhaimeid and Martin, 2020). Moreover, this approach was applied in the AAPM TG-293 as an alternative approach to the use of CT numbers from actual images (AAPM, 2019b). The alternative approach could be, sometimes, a solution to overcome the difficulty of getting a clinical access to actual images due to patient privacy requirements. Therefore, mean energies of the 80–140 kVp spectra, i.e. monoenergetic x-ray beams, were used in this study to calculate $CT(x, y)$ for each pixel of the ROI, and then $\overline{CT}(x, y)_{ROI}$ as in Eq. (6).

The beam quality is measured in terms of half values layer (HVL) and was measured for different CT scanners using various scan parameters and found to be 5.4–9.7 mm of aluminium (Mathieu et al., 2010). The HVL values were used to estimate the mean energies for the 80–140 kVp spectra using a MATLAB code called SPEKTR (V3.0) (Punnoose et al., 2016), and found to be in the range of about 50–72 keV. Therefore, five mean energies (50, 60, 70, 80, 90) keV were used in this study to assess D_w for each slice within all the phantoms. This is to cover the typical spectral range used for CT scans and allow

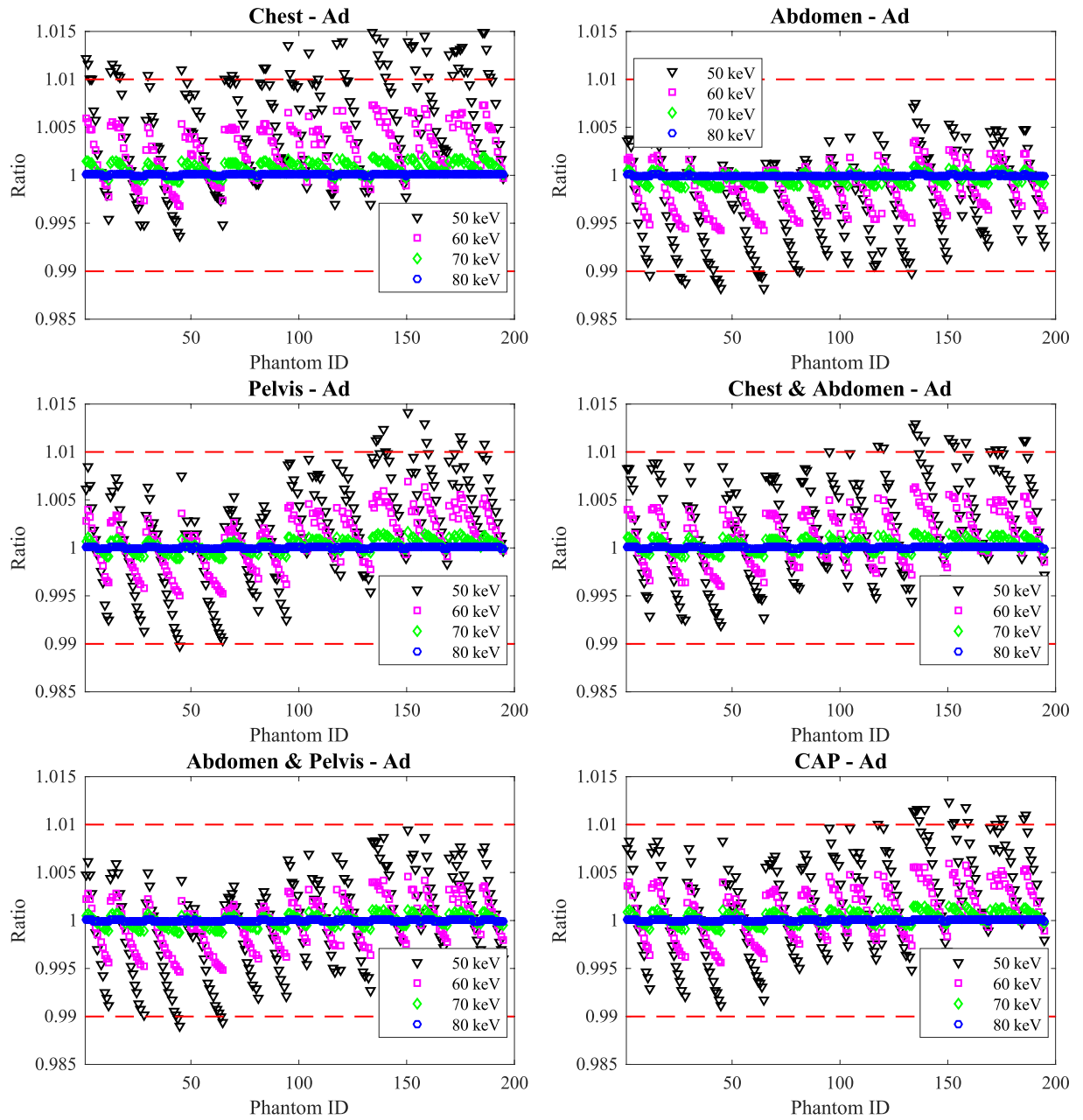


Fig. 5. Ratios of D_w calculated for the six regions of the trunk for the paediatric (Ad) phantoms with various mean energies and normalized with respect to those for 90 keV.

the energy variations among patients and the beam hardening effect to be considered (Li et al., 2012). $CT(x, y)$ was assessed for each voxel within the phantoms as in Eq. (4) using the density and the chemical composition of each voxel provided within the phantoms library, and the mass attenuation coefficients measured by the National Institute of Standards and Technology (NIST) for the energies investigated to calculate the effective $\mu_{(x,y)}$ for each voxel (NIST, 2020). This allowed calculation of $CT(x, y)_{ROI}$, and hence A_w and D_w for the regions studied. The D_w calculation method used in this study was similar to that reported in AAPM TG-293 (AAPM, 2019b).

3. Results and discussion

3.1. D_w values for the phantoms

D_w values calculated with the different mean energies for the head

region are plotted against ED values in Fig. 1. The results show general trends, but plots are disjointed as several different models for the head were used for each age in the library with slight differences in relative dimensions (Fig. 1). Results suggest that the x-ray spectrum plays a significant role in determining values of D_w for the head in paediatric and adult patients. Ratios of D_w/ED vary by factors of 1–1.09 for paediatric patients and 1–1.10 for adults, with the majority of the change occurring for the 50 and 60 keV energies, and that between 70, 80 and 90 keV being less than 1.01. The D_w values increase with the size of the patient for the different keV values. The differences were 1.03 for the smallest paediatric patient and increased up to 1.09 for the largest paediatric patient. There was a similar trend for adult patients, where the variations increased with size of the patient from a factor of 1.06–1.10.

D_w values calculated with the different mean energies for the trunk regions are plotted against ED values in Fig. 2 and 3. In contrast to the

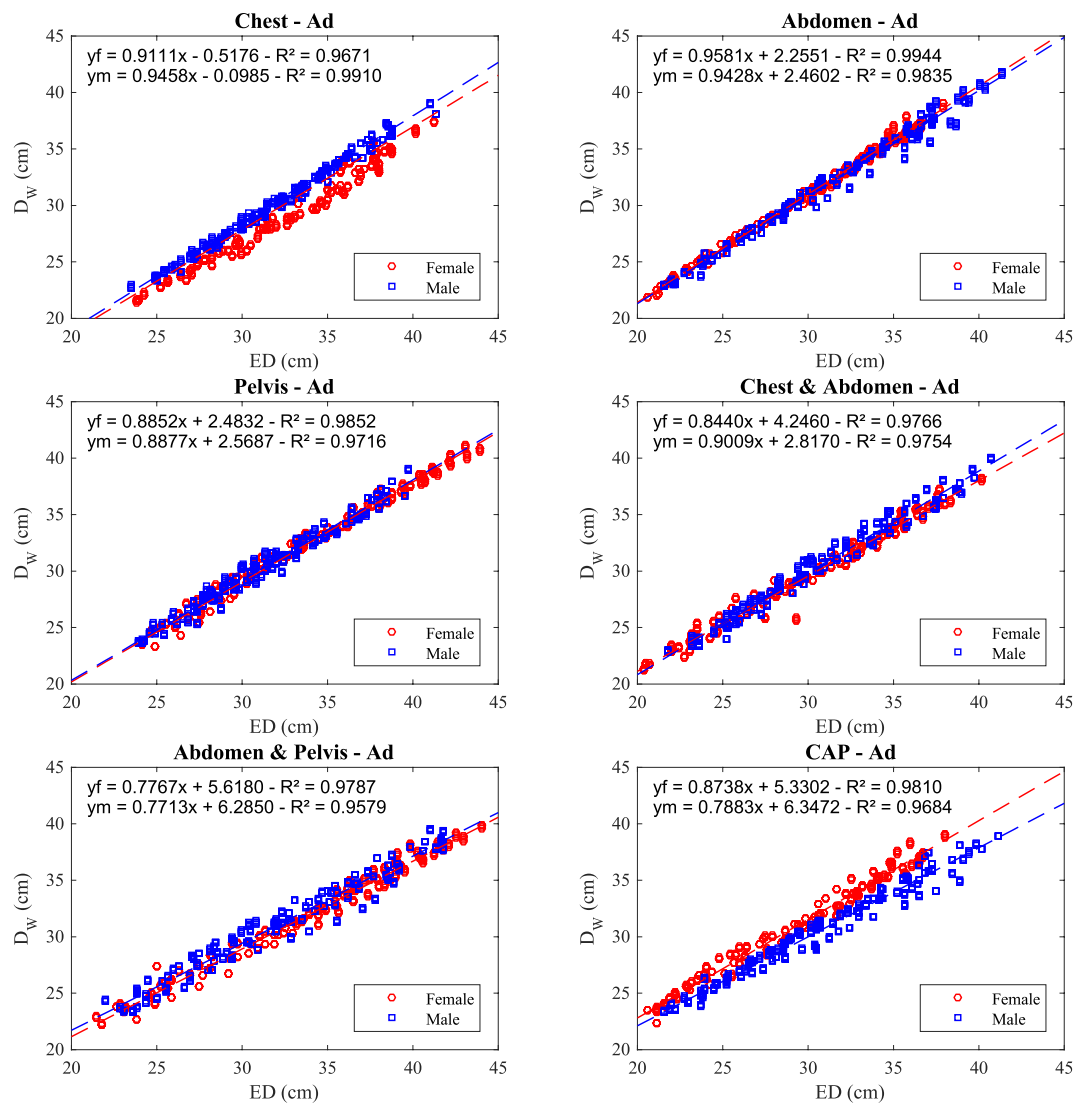


Fig. 6. Gender-specific correlations between ED and D_w values for the adult phantoms (Ad) assessed in the trunk regions. The linear regressions (yf) and (ym) are for the female and male phantoms, respectively.

head region, variations between D_w values calculated with the different mean energies for the trunk of the paediatric and adult phantoms were minimal. In order to demonstrate the influence of differences in the x-ray spectra more clearly, D_w values that were calculated in regions of the trunk with the mean energies of 50–80 keV have been normalized with respect to those of the largest energy 90 keV and plotted in Fig. 4 and 5. Almost all the variations were within a factor of ± 1.01 . D_w values calculated with 70 and 80 keV were almost equal to those for 90 keV, and even values for 50 and 60 keV only extended to ± 1.015 . In addition, the size of the patients had a negligible impact on the variations.

The large variations found between D_w values for the different beam energies in the head region resulted from high density areas caused by skull, which increased with patient size as shown in Fig. 1. However, the trunk regions that began just below the neck were not affected significantly by high density tissues as the majority of the trunk is made up from soft tissues. Some effects, were observed at the lower energy 50 keV, in the chest and pelvis regions due to influence of the bone tissues in these regions that absorb low energy photons. The effects resulted from ribs and the shoulder that is included in the chest region as well as the pelvic bone in the pelvis region. These effects, however, were not as significant as in the head region, where the skull bone becomes the predominant tissue at the upper and lower ends.

The variations found between D_w values calculated with 60 and 80 keV in the adult trunks (Fig. 5) were in good agreement with those reported in a previous study by (Li et al., 2012). They calculated D_w values for six different patient sizes in the various regions in the trunk using two mean energies 62 and 83 keV for an x-ray spectrum of 120 kVp. The two mean energies were used to account for the beam hardening effect that would occur when an obese patient was scanned. The D_w values calculated with the two energies were all within $\pm 1\%$ for the different regions.

3.2. Correlations of ED and D_w values

Although assessment of D_w based on CT images by the method discussed in the AAPM TG-220 should be considered as the accurate method, it requires advanced software integrated within the scanner system. However, such software is not available in CT scanners at the present time, and most studies are conducted retrospectively. Therefore, use of the linear regressions of the correlations given in Figs. 2 and 3 for the trunk regions could provide a simple and practical approach to assessment of D_w and SSDE for any given patient size. The variations found in D_w values calculated in the trunk with the different mean energies that represent the typical range used for CT scans are minimal, all within $\pm 1.5\%$. Thus, correlations between ED values and

Table 1

Evaluation of using of the linear regressions for the correlations shown in Figs. 2 and 3. ED and $D_{w,AAPM}$ values were measured from CT images and presented in (AAPM, 2014). D_w values were assessed in this study using the same ED values.

	Phantom size	ED (cm)	$D_{w,AAPM}$ (cm)	D_w (cm)	Difference (%) ⁽¹⁾
Abdominal phantoms					
Water Cylinder	20 cm	20.0	20.4	20.9	2.5
	30 cm	30.0	30.6	30.9	0.9
Abdomen Water Phantom	15 cm	13.3	13.5	14.1	4.8
	20 cm	17.8	18.0	18.7	3.8
	25 cm	22.3	22.7	23.2	2.3
	30 cm	26.7	27.2	27.8	2.1
	35 cm	31.2	31.6	32.0	1.4
	40 cm	35.6	36.1	36.2	0.4
	45 cm	40.1	40.5	40.5	0.0
Tissue Equivalent Abdomen Phantom	Newborn	10.1	10.2	10.9	7.0
	1 Yr	13.1	13.3	13.9	4.9
	5 Yr	16.5	16.7	17.4	4.1
	10 Yr	18.8	19.1	19.7	3.1
	15 Yr	21.9	22.3	22.8	2.4
	Small Adult	26.8	27.2	27.8	2.3
	Medium Adult	29.6	29.8	30.5	2.4
	Large Adult	36.0	36.6	36.6	0.1
Thorax Phantoms					
Tissue Equivalent Thorax Phantom	30 cm	25.4	20.9	23.3	11.5
	35 cm	30.7	27.3	28.2	3.3
	40 cm	35.7	33.2	32.8	-1.3
Anthropomorphic Phantom		28.7	25.6	26.3	2.9
				Average	2.9

(1) Difference (%) = $(D_w - D_{w,AAPM}/D_{w,AAPM}) \times 100$.

energy-independent D_w values could be established. R^2 values for the correlations were in the range of 0.96–0.99 for paediatric and 0.94–0.99 for adults.

The correlations were affected slightly by gender, particularly in the chest region, which subsequently had an influence on other scanned regions that included the chest. For example, the breast and shoulder sizes in males and females are different as well as the overall shape in the trunk. Fig. 6 shows the correlations based on gender for the regions studied, and the R^2 values for the chest and CAP are slightly higher than the gender averaged data. Therefore, if gender-specific correlations were used, this would improve the precision of the assessment of D_w based on the ED slightly. Nevertheless, the correlation that can be achieved with a gender neutral approach is good. Implementation of the correlations based on either gender neutral or gender-specific factors could be accomplished readily at the present time, as it is relatively independent of beam energy. Even when software systems to calculate SSDE are integrated into new CT scanner models, this approach might be useful for application to older models that could not be upgraded.

The AAPM TG-220 report presents a comparison between ED and D_w values conducted on CT images of various phantom sizes by (Wang, Christner, et al., 2012). The ED values given in the report were used to assess D_w values using the linear regression of the correlations shown in Figs. 2 and 3, which are based on values for both genders. Table 1 gives comparisons between these D_w values and those in the AAPM report ($D_{w,AAPM}$) for different sizes of abdominal and thorax phantoms. The comparisons allow the accuracy of the correlations to be evaluated and an indication of the uncertainty that would result from use of an energy-independent approach. The differences between $D_{w,AAPM}$ values and those assessed in this study were less than 5%, except for two values, for the newborn abdomen and the smallest thorax phantom. Nearly, all the values were slightly lower and the average difference was 2.9%. This indicates that this approach might be considered as a simple energy-independent method to estimate D_w and hence SSDE with a reasonable level of accuracy.

Two best-fit curves for the body conversion factors f_{size}^{32cm} were given in the AAPM TG-204 report and used in the TG-220 report, one for 120 kVp and the second for all tube potentials 80–140 kVp. It has been

found that the average percent difference of the absolute differences for factors assessed using the two curves was 5.1% over all sizes (AAPM, 2011, 2014). This difference dropped to 4.3% when only sizes of 10–32 cm were included, and further declined to 3.1% when only the more common sizes seen in the clinic (15–32 cm) were considered. Therefore, a single set of factors that were based on 120 kV were suggested for use with all tube potentials due to the small differences. In this investigation, the percentage absolute differences between the patient sizes over the different tube potentials studied were 0.01–1.54%, 0.18% on average, for paediatric patients, and 0.01–1.52%, 0.20% on average, for adults. Therefore, the uncertainty in the SSDE calculation would be the combined average uncertainties that result from determination of the patient size and calculation of f_{size}^{32cm} using Eq. (2). Since the average differences found in this study were small compared to the uncertainty in the SSDE calculation based on f_{size}^{32cm} , they would only raise the overall uncertainty by 0.01%, so the average uncertainties would be similar to those reported by the AAPM. For example, the uncertainty of SSDE for adults would be 3.11%.

4. Conclusions

ED and D_w were assessed for paediatric and adult phantoms of a wide range of sizes from a small baby to adult ages over seven scanned regions. Five mean energies that represent an x-ray spectra range of 80–140 kVp were applied to take the beam hardening effect into consideration. The x-ray spectrum was shown to affect D_w values in the head region significantly. Values for D_w assessed with the different energies varied by factors of 1–1.09 for paediatric and 1–1.10 for adult. D_w values of the trunk regions, on the other hand, were only influenced slightly by the x-ray spectrum, where almost all the D_w values assessed with the different energies were within a factor of ± 1.01 . This allows energy-independent correlations to be established between ED and D_w values for the trunk regions. The correlations can be easily implemented without the need for advanced software to assess D_w values. Comparisons with values given in the AAPM report calculated by a different method showed differences of less than 5% for most results, with an average of 2.9%.

Author statement

Abdullah Abuhaimeid: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing - Original Draft.

Colin J. Martin: Methodology, Investigation, Writing - Review & Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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